NICHOLAS GENOVESE

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EDUCATION

University of Pennsylvania, School of Engineering, Philadelphia, PA

MSE in Mechatronics and Robotic Systems (Accelerated Masters Program) — **GPA: 3.80/4.00** — **May 2022**

BSE in Mechanical Engineering — **GPA: 3.45/4.00** — **May 2021**

Minor in Engineering Entrepreneurship, Graduated Cum Laude

Stanford University BioDesign Executive Education Program — **Mar 2024**

Managing Innovation 2024

Geometric Dimensioning and Tolerancing Certification — **Nov 2023**

Application and Interpretation of ANSI/ASME Y14.5 GD&T

Lean Six Sigma Black Belt Certification - Design Excellence — **July 2023**

Six Sigma Solutions and Johnson and Johnson Supply Chain Academy

SUMMARY

Senior R&D Engineer at Johnson & Johnson MedTech with a Masters degree from UPenn in Mechatronics and Robotic Systems and proven expertise in robotics, automation, and complex hardware system development. Skilled at designing and scaling electromechanical systems with a focus on rapid prototyping and cross-functional leadership. Passionate about developing innovative robotic solutions.

SKILLS

Engineering: Circuit Design and Assembly, 3D Printing, Fixture Development, Manufacturing System Design, Mechanical Testing

Modeling/Software: SolidWorks, MATLAB, Minitab, C++, Python, MPLAB X IDE, ANSYS (FEA)

PROFESSIONAL EXPERIENCE

Johnson & Johnson, Raritan, New Jersey

Senior Engineer, Project Lead: Innovation and Partnerships — **Feb 2025 – Present**

- Lead R&D development of a novel single-use Negative Pressure Wound Therapy wearable device projected >$560MM NPV
- Designed and implemented a semi-automated robotic manufacturing assembly process to reduce prototype build time by 50%
- Innovate line extensions in Robotic Microsurgery platform to expand projected global revenue from ~$30MM to ~$100MM
- Inventor on 15+ IP Submissions, 3 patent pending related to various technical projects

R&D Leadership Development Program: — **Graduated: Feb 2025**

J&J Early-in-Career 3-year Accelerated Rotational Program

Senior Engineer, Technical Lead: Interventional Oncology — **Oct 2023 - Feb 2025**

- Developed and executed internal Factory Acceptance test plan to validate $600k NIR Fiberscope hardware system
- Optimized dose control of cryoablation therapy administration in preclinical studies by over 40% through internal R&D efforts
- Managed 15 external engineers in development of localized chemo delivery device integral for Phase 1b Clinical Trial

R&D Engineer, Technical Lead: Ethicon WC&H — **June 2022 - Oct 2023**

- Achieved accelerated 3 month internal approval regarding design efforts for wearable wound closure device
- Leveraged SolidWorks and J&J MakerSpace to innovate novel design prototypes for preclinical evaluation
- Developed, validated, and implemented testing protocols for FDA 510k submission package of wearable wound closure device

J&J DEI & Culture Team: Led fundraising efforts of over $17,000 across multiple J&J and American Lung Association events

United States Naval Surface Warfare Center, Philadelphia, PA

NRIEP Intern, US Navy Code 416 Team - Submarine Life Support Systems — **May 2021 – Aug 2021**

- Developed and tested functional life support systems for implementation in Naval submarines
- Collaborated on Manned to Unmanned Vessel design projects involving automated naval vessel deployment

LEADERSHIP & PROFESSIONAL DEVELOPMENT

Director of Fundraising, Zandrability Inc.

http://zandrability.com/ — **Sept 2021 – Present**

- Promote all fundraising efforts towards finding a cure for Spinal Muscular Atrophy (SMA)

University of Pennsylvania, Philadelphia, PA

Teaching Assistant: Mechanical Engineering Lab 248-249 — **Sept 2021 – May 2022**

- Taught a weekly lab session to third year undergraduate Mechanical Engineering students

Senior Design Project: Self-Sustaining UV Water Purification System for Kumasi, Ghana — **Sept 2020 – May 2021**

- Awards: William K. Gemill Memorial Prize for Outstanding Creativity, Leadership Award
- Placed Top 3 out of 100+ Senior Projects in UPenn School of Engineering Senior Design Competition

Aerial Robotics Club: Mechanical team member designing and developing autonomous aerial systems — **Jan 2018 – May 2021**